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Evolution of satellite mobile communications

Whitepaper

Abstract:

Satellite communication services are perceived to be for defense applications, weather forecasting, and media broadcasting. This paper traces the evolution of satellite mobile communications for commercial voice and broadband services. It is also shown how satellite mobile technologies have been derived from terrestrial mobile standards and are becoming mainstream with 5G standardization.

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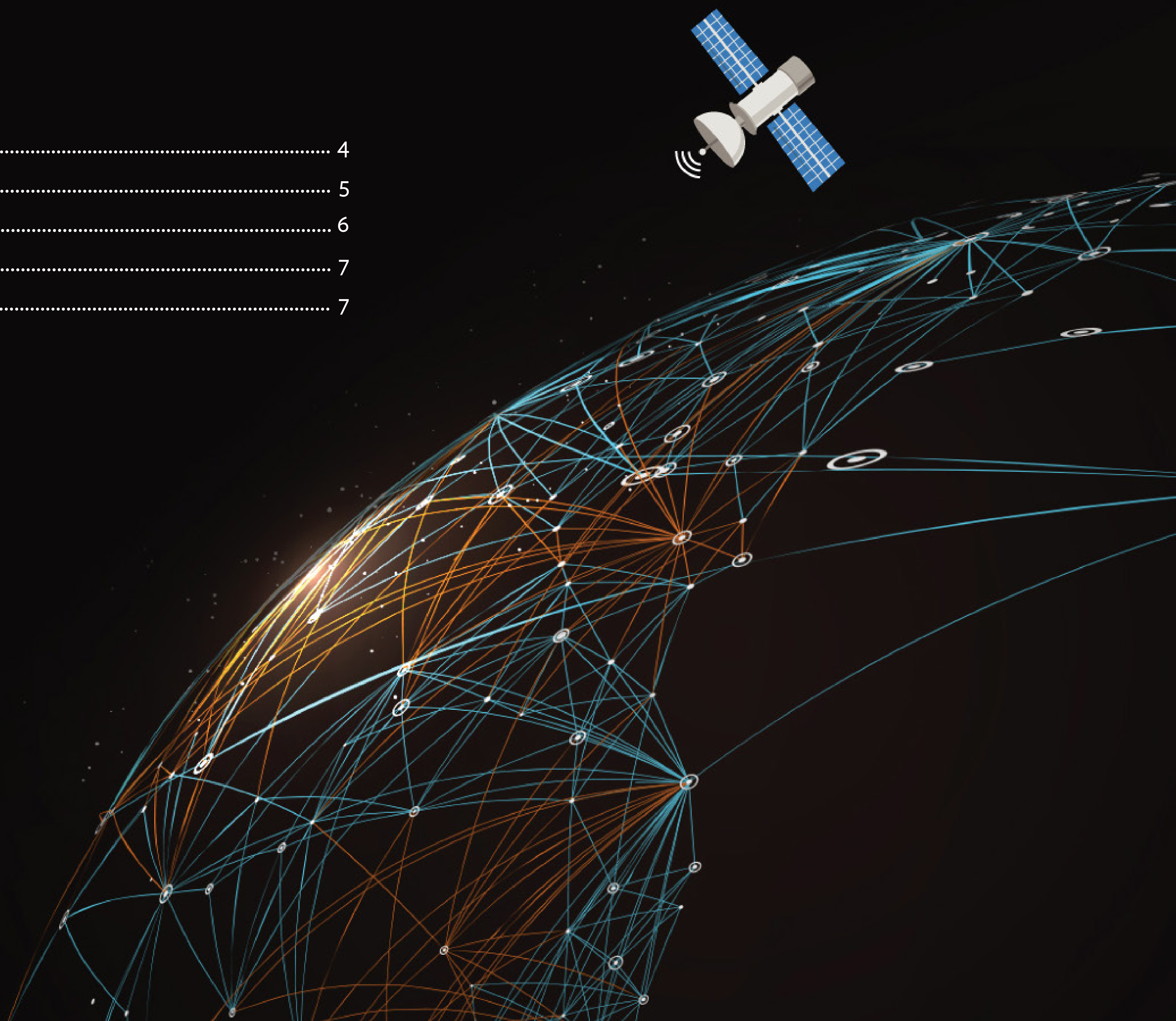

The beginnings

Satellite communications are a variant of basic wireless telecommunication. The SHF band of 3GHz to 30GHz frequencies is used to transmit electromagnetic waves that pass through the earth's ionosphere to reach satellites in space. Passive satellites relay the signals back to receivers on the ground or in the air. Active satellites process and retransmit signals to receivers, often changing the frequency of transmission.

Arthur C. Clarke's 1945 article "Extra-terrestrial relays: Can rocket stations give worldwide radio coverage?" launched pioneering work in satellite communication. Clarke concluded that worldwide coverage could be achieved with just three geostationary satellites. He also predicted TV broadcasts that transcend national borders and achieve worldwide coverage. Clarke also anticipated the use of solar energy for satellite operation.

From the launch of the aluminized balloon satellite Echo-1 in August 1960, satellite systems emerged from experiments to military use during the 1960s. Three geostationary satellites would cover the entire globe. But to balance the Earth's gravitational pull, these satellites would have to be 36,000 km from Earth. This would mean high transmit power and one-way delays exceeding four hundred milliseconds. Medium and low earth orbits would be alternate choices where a greater number of revolving satellites is required. In these, lower power and delays are achieved at the cost of increased complexity for tracking and receiving from fast moving satellites.

By 1970, satellite communications had proven viable for public domain applications such as weather forecasting, TV and radio broadcasting, and communication with ocean-going ships. Inmarsat's maritime communication services were hugely popular. This was followed by asset tracking services such as Omnitracs.

Mobile satellite services

Public mobile satellite communications were heralded by the announcement of the Iridium system in 1990. The objective was to provide worldwide mobile services via a constellation of sixty-six low-earth orbit satellites. Globalstar was a competing system based on geostationary satellites. These systems went live in the early 2000's, making satellite mobile communications available for commercial use. The services were voice-based.

This was coincident with the emergence of 3G technologies in terrestrial networks for mobile data. We will follow the evolution of satellite communications from mobile voice to broadband data with examples from the worldwide satellite communications service provider Inmarsat.

A pioneer in maritime communication services, Inmarsat launched the Broadband Global Access Network (BGAN) service in 2005. The technology was 3G-based and up to 492 Kbps was being delivered to individual terminals. Voice was supported as Voice-over-IP.

This was followed by the Global Satellite Phone Service (GSPS) in 2009. The technology used by Inmarsat was the GSM-based GMR1-2+. Circuit switched data at 2.4 bps was supported. Handheld phones and data modules were available.

In 2016, Inmarsat launched the higher-end Global Xpress (GX) service. To achieve broadband rates of 50 Mbps download and 5 Mbps upload, DVB-S2X technology was used. This is an evolved version of the popular digital broadcast standard DVB-S2. At the highest end of satellite broadband, ViaSat and EchoStar deliver throughputs exceeding 1 Gbps.

As in the case of 2G and 3G wireless technologies, 4G LTE has caught the attention of satellite service providers. Since 2015, this global mobile standard has been chosen by several companies and governments for deriving proprietary variants deployable over satellites. Today, there are several LTE-based satellite communication networks, some of which are in the proof-of-concept stage while others are undergoing advanced deployment trials. These include government-funded critical communication networks, and commercial networks.

Unlike 4G LTE, which is deployed via proprietary adaptations for satellites, satellite support is becoming mainstream in 5G standardization. 3GPP, the global standards body for mobile broadband, is evolving satellite variants for its 5G NR specifications. This will provide open standards which can be adopted by any satellite service provider. Terrestrial 5G phones with a satellite mode option are also a possibility with this.

Conclusion

In the 1960s, satellite communications emerged as a vital component for defense communications and weather forecasting. Services have evolved to include TV broadcasting, maritime services, and asset tracking.

Satellite mobile services followed on the footsteps of terrestrial mobile networks. From voice-centric services in the early 2000s, data services followed. Data bandwidths started as low as 2.4Kbps and have scaled beyond 1 Gbps.

Terrestrial 2G/3G/4G have been adapted for use in the satellite segment. With increased use cases for satellite mobile data, satellite communication has become an essential component in 5G standardization efforts.

With more than two decades of product engineering record in satellite communications, Sasken is actively involved in the convergence of satellite and terrestrial technologies for wireless broadband. Sasken has delivered in-house designed handheld devices and ruggedized terminals for a variety of customers. From being a successful IP and services provider on 3GPP based terrestrial networks, Sasken has scaled to offer bespoke product engineering services for satellite devices and access networks.

About the Author

Sharansunil Lekkad is an industry veteran whose tenure at Sasken has spanned more than two decades. Sharan had managed some of the most complex products engineered by Sasken, including satellite phones and modules deployed worldwide. Sharan is currently engaged in proof-of-concepts and pre-commercial trails of two LTE-based satellite networks and is an avid follower of trends in the satellite mobile communications market. He has managed programs involving full device development, productization, manufacturing support, and end-of-life component management. Sharan remains a favorite mentor for newer generations of engineers at Sasken.

About Sasken

Sasken is a specialist in Product Engineering and Digital Transformation providing concept-to market, chip-to-cognition R&D services to global leaders in Semiconductor, Automotive, Industrials, Consumer Electronics, Enterprise Devices, SatCom, Telecom, and Transportation industries. For over 32 years and with multiple patents, Sasken has transformed the businesses of 100+ Fortune 500 companies, powering more than a billion devices through its services and IP.

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